

Title

Feasibility Assessment and Protocol for Comparing Two Empirical Broad-Spectrum Antibiotics in Septic Critically-Ill Patients: The EMPRESS Trial Design

Abstract

Sepsis management in intensive care units relies upon rapid empirical antimicrobial therapy, but comparative evidence regarding optimal antibiotic selection remains limited. The EMPRESS trial will conduct an integrated feasibility evaluation of a subsequent definitive trial comparing two commonly-used broad-spectrum agents in critically ill septic patients. An initial feasibility phase will enrol 200 participants across 10 intensive care units and assess five pre-defined feasibility criteria: recruitment pace, data completeness, protocol adherence, participant retention, and outcome measurement timeliness. Eligible critically ill adults with sepsis will be randomly assigned in a 1:1 ratio to receive either meropenem or piperacillin/tazobactam as empirical therapy. The primary endpoint will be all-cause mortality at 30 days post-randomisation. Eligibility criteria will specify organ dysfunction severity and recent antimicrobial exposure to ensure comparable baseline risk. Secondary safety outcomes will track serious adverse reactions. An interim feasibility review will determine whether the definitive trial should proceed unchanged or with protocol modifications.